

Gravitational Reference Frames in Cortical Networks

Multisensory Orientation, Embodied Neural Rendering, and Tests of Local, Broadcast, and Recurrent-Distributed Architectures

Micah Blumberg

Self Aware Networks Research Institute

<https://selfawarenetworks.com>

August 13, 2026

Preprint; not peer reviewed.

Abstract

A room ordinarily remains upright while the eyes, head, and body move, even though retinal coordinates rotate and each sensory cue is uncertain. Macaque electrophysiology identifies gravity-aligned and retina-aligned object information, mixed parietal reference frames, and gravity-aligned scene structure. These findings establish a representation problem but do not reveal whether participating populations estimate gravitational direction locally, receive a specialist broadcast, or constrain it through recurrent exchange. This paper formalizes those three architecture families using circular cue likelihoods, typed coordinate transforms, response tuning, transfer, residual covariance, perturbation spread, propagation delay, and adaptive cue reweighting. A constructive nonidentifiability result shows why unrestricted static observations cannot select among them. The executable program reproduces the published gravity/retina partitions from released arrays, evaluates pure and mixed synthetic mechanisms under missingness, hidden-node, nuisance, topology, and distribution shifts, and machine-checks fourteen finite reporting and nonidentifiability theorems. The strongest synthetic evaluation compares four fully trainable recurrent models with identical 291-parameter cores and different local, broadcast, graph, or joined observation views. Across 37,632 prediction rows, graph has the lowest aggregate error among the capacity-matched models at 0.1671 with 95% seed-cluster interval [0.1637, 0.1712] and coverage 0.8523. A historical fixed reservoir has descriptively lower error, 0.1544, but only 0.2119 coverage. Held-out topology reverses the graph-versus-broadcast ordering; combined shift breaks nominal calibration; model disagreement approaches one; and 5,376 intervention rows retain 1,728 exact structural nulls and 826 negative changes or reversals. The result is a bounded measurement program. It supports time-resolved, topology-aware, perturbation-bearing experiments with explicit nuisance models and refusal states. It does not identify a biological cortical architecture, equate a neural estimate with physical gravity, establish conscious rendering, or prove Self-Aware Networks as a complete theory. The biological final test remains sealed.

Keywords: gravitational reference frame; multisensory integration; coordinate transformation; inferotemporal cortex; vestibular system; embodied perception; recurrent computation; neural rendering; causal identifiability; Self-Aware Networks

1. The ordinary problem comes first

Look at a doorway, tilt your head, and then stand upright again. The retinal image of the doorway rotates, but the doorway usually does not seem to fall sideways. Reach toward a cup while

leaning, and the nervous system must relate the cup, the hand, the head, the support surface, and gravitational vertical quickly enough to guide action. In virtual reality, a disagreement among these relations can produce instability or motion sickness even when every displayed pixel is internally consistent.

This apparently simple stability hides several computations:

1. the eyes encode an image relative to the retina;
2. eye position changes the relation between retinal and head-centered coordinates;
3. vestibular signals constrain head motion and orientation;
4. somatosensory signals constrain posture, support, and body configuration;
5. visual horizons, surfaces, and object regularities supply environmental orientation cues;
6. motor commands and their sensory consequences update the estimate; and
7. different objects and actions must use the result without requiring a complete reconstruction from the beginning on every occasion.

The established scientific vocabulary includes reference frames, coordinate transformations, multisensory cue integration, population coding, gain fields, attractor dynamics, recurrent processing, and reliability-dependent reweighting. These terms already explain important parts of the problem. The residual question addressed here is more specific: **what network architecture makes a gravity-aligned relation available to multiple representations and actions?**

Only after that problem is clear do we need a compact name. This paper calls the usable neural relation a **gravitational reference frame in cortical networks**. The phrase is operational: it refers to a time-varying estimate that allows neural responses, percepts, or actions to remain systematically related to gravity when retinal, head, body, and environmental cues change. It is not a claim that gravity is computed by one unique cortical location, that every cortical column contains the same variable, or that neural signals are themselves a new physical theory of gravity.

2. What the evidence already establishes

2.1 Gravity-aligned object orientation in anterior inferotemporal cortex

Emonds et al. recorded orientation-selective neurons in macaque anterior inferotemporal cortex while animals viewed objects under body tilt and visual-ground manipulations [1]. With body tilt of approximately ± 25 degrees, retinal orientation changed by about 40 degrees. Among neurons with consistent orientation tuning in the full-scene condition, the released arrays contain 33 significant only in gravitational coordinates, 11 only in retinal coordinates, and 8 significant in both. With an isolated object, the corresponding partition is 32/14/7. Under visual-ground cue conflict, the partition becomes 43/67/13. These are not three mutually exclusive biological cell types in an absolute sense; they are analysis categories under specified conditions and significance procedures.

The important point is affirmative. Anterior inferotemporal activity can contain orientation information aligned with gravity, retinal orientation, or both. Visual background cues as well as vestibular or somatosensory information contribute. A population decoder can read useful orientation information from these responses [1].

The result does not specify whether each recorded population independently estimates gravitational direction, receives it from another circuit, or participates in recurrent estimation. It measures the

coordinate relation expressed in responses, not the full causal route that produced it.

2.2 Mixed and region-specific reference frames

Rosenberg and Angelaki found a heterogeneous continuum of object-tilt representations in caudal intraparietal cortex, ranging from egocentric to intermediate and gravity-centered forms [2]. Vaziri and Connor found gravity-aligned scene-structure coding in the ventral pathway [3]. Chen, DeAngelis, and Angelaki documented diverse vestibular reference frames across parietal areas rather than one uniform coordinate system [4]. Caruso and colleagues further showed that visual and auditory reference frames can change over time and differ across regions as signals approach action-relevant alignment [5].

These findings make a static, all-or-nothing description inadequate. A frame may be mixed, task dependent, region specific, temporally evolving, and readable at the population level even when no individual unit represents a complete transformation. That complexity is not noise to be discarded. It supplies the observables needed to distinguish mechanisms.

2.3 Physical stability, verticality, and cue reliability

Human fMRI work identifies generalizable information about physical stability in a frontoparietal network under tested conditions [6]. Psychophysics and lesion evidence show that head tilt, body tilt, vestibular pathways, and cortical vestibular regions influence perceived verticality and mental rotation [10-13,15]. Cue-integration models show how populations can combine uncertain signals [7,8], while Fetsch et al. demonstrated rapid visual-vestibular reweighting as reliability changed in human and monkey self-motion judgments [9]. Head-direction research provides a complementary example in which vestibular and other self-motion signals update a persistent orientation estimate [16].

Together, this literature establishes a distributed and adaptive scientific problem. It does not force a choice among the three architectures introduced below. Rather, it constrains any viable model: the model must permit mixed frames, cue conflict, reweighting, temporal evolution, partial regional specialization, and action-dependent use.

3. Source genealogy and the exact 2026 contribution

This paper is not presented as if its entire formulation appeared at once. The historical record contains stages with different evidentiary strength and conceptual completeness.

3.1 Distributed multisensory prediction, 2011-2012

An author note internally dated May 13, 2011 describes a stereographic-camera analogy in which distributed neural "pixels" contribute different portions of a changing scene; learned statistical relations support inference and prediction; and multiple senses contribute to one continuously updated picture [19]. The 2011 date is an owner-supplied date. The exact text is publicly fixed in Git on June 8, 2022, so the two provenance facts must remain distinct.

An August 2012 public recording describes the mind as an internal projector that predicts the world, with external input constraining a larger internally supplied construction [20]. These sources establish early distributed-prediction and rendering ancestry. They do not yet define a gravity vector, choose a cortical architecture, or contain the equations proposed here.

3.2 Spatial computing and a time-varying neural field, 2017

The April 11, 2017 Neural Lace Podcast explicitly describes the brain in terms of spatial computing, distributed feature-to-object inference, and a time-varying three-dimensional ionic pattern, while connecting the account to future read/write brain-computer interfaces [21]. This is a stronger bridge to the present problem because a stable spatial model must survive changing sensor and body coordinates. It still does not contain the current local/broadcast/recurrent discriminator.

3.3 The local-or-transmitted question, 2024

The publicly Git-fixed January 2024 *Self Aware Networks* source takes in the Emonds et al. result and asks whether gravitational reference information is calculated in each cortical reference frame or transmitted among participating cortical populations [22]. A January 24, 2024 SVGN article supplies a broader traffic motif involving local laminar processing, corticothalamic exchange, and long-range feedforward and feedback pathways [23]. The latter is used here as a historical bridge, not as authority for every anatomical sentence in that article. In particular, inferotemporal cortex and intratelencephalic neurons share the abbreviation "IT" but are not the same anatomical entity.

3.4 Additions made in this manuscript

The following are 2026 contributions and are not backdated:

- explicit local, broadcast, and recurrent-distributed candidate architectures;
- a circular-statistical formalization of the shared estimation problem;
- a constructive demonstration of static observational nonidentifiability;
- exact reproduction of bounded results from released data arrays;
- a known-ground-truth synthetic discriminator benchmark;
- perturbation, timing, transfer, and adaptation endpoints; and
- experiment designs that can select, combine, or reject the idealized models.

This staged genealogy preserves the continuity of the research without converting an early conceptual ancestor into a claim that later mathematics or experiments had already been completed.

4. Variables and coordinate relations

Let an object feature be represented by a vector x . Superscripts identify coordinate systems: eye/retinal E , head H , body B , and gravity/world G . A simplified coordinate chain is

$$x^G(t) = R_B^G(t)R_H^B(t)R_E^H(t)x^E(t), \quad (1)$$

where each R is a rotation between coordinate systems. This equation is a bookkeeping idealization. Biological populations can implement transformations implicitly through tuning, gain modulation, basis functions, recurrence, or learned sensorimotor relations rather than multiplying explicit matrices [7,8]. Translation, perspective, articulated body geometry, and uncertainty can be added when the task requires them.

For orientation in a plane, let $g(t) \in [-\pi, \pi)$ denote the latent gravitational direction relevant to the task. Let cue m provide circular observation $q_m(t)$, with concentration or reliability $\kappa_m(t)$. A von Mises likelihood is

$$p(q_m(t) | g(t), \kappa_m(t)) = \frac{\exp\{\kappa_m(t) \cos[q_m(t) - g(t)]\}}{2\pi I_0(\kappa_m(t))}, \quad (2)$$

where I_0 is the modified Bessel function of order zero. The cue may be visual, vestibular, somatosensory, motor-predictive, or memory-derived. High κ_m indicates a concentrated cue; low κ_m indicates uncertainty.

A reliability-weighted circular estimate can be written

$$\hat{g}(t) = \arg\left(\sum_{m=1}^M w_m(t) e^{iq_m(t)}\right), \quad w_m(t) = \frac{\kappa_m(t)}{\sum_{n=1}^M \kappa_n(t)}. \quad (3)$$

Equation 3 is not asserted as the unique neural algorithm. It is a common computational target against which biological and alternative models can be tested. Non-Bayesian biases, correlated noise, priors, causal inference, and task-dependent weights may all matter.

5. Three architectures that can produce similar static responses

5.1 Model L: local estimation

In the local model, each participating population j estimates a useful gravitational relation from its own available cues and parameters:

$$\hat{g}_j^L(t) = F_j(q_{j1}(t), \dots, q_{jM_j}(t); \theta_j). \quad (4)$$

The functions F_j can differ by modality, region, layer, or task. Similar estimates can emerge because populations receive similar cues or learn under the same environment, not because one population transmits a common variable. Under a focal perturbation, the directly affected population changes first; other populations change only through ordinary downstream consequences, shared new sensory evidence, or later learning.

5.2 Model B: specialist computation and broadcast

In the broadcast model, one or more specialist circuits integrate cues into a shared latent estimate $z(t)$, which is distributed to receivers:

$$z(t) = F_B(q_1(t), \dots, q_M(t); \theta_B), \quad \hat{g}_j^B(t) = H_j(z(t), u_j(t); \phi_j), \quad (5)$$

where $u_j(t)$ contains receiver-specific input. "Broadcast" need not mean a single anatomical nucleus or simultaneous arrival at every neuron. It means that a low-dimensional upstream variable contributes a common source of orientation information across receivers. Perturbing the source should produce comparatively rapid, widespread, source-locked effects, conditioned on connectivity and conduction delays.

5.3 Model R: recurrent-distributed constraint

In the recurrent-distributed model, population estimates are updated by local cues and exchanges with other populations:

$$g_j^R(t + \Delta t) = \text{wrap} \left[(1 - \alpha_j - \beta_j) g_j^R(t) + \alpha_j F_j(q_j(t)) + \beta_j \sum_{k=1}^N A_{jk} g_k^R(t) \right], \quad (6)$$

where A_{jk} is a directed, normalized coupling matrix, α_j weights fresh local evidence, and β_j weights recurrent constraints. More realistic versions would use circular vector sums, delays, nonlinearities, and uncertainty. The defining prediction is not simply that feedback exists. It is that a usable shared relation emerges through time- and topology-dependent mutual constraint.

5.4 Neural response readout

For a neuron or population with preferred orientation μ_i , response tuning under candidate estimate $\hat{g}_j$ can be represented as

$$r_i(t) = b_i + a_i \cos(o^E(t) - \hat{g}_j(t) - \mu_i) + \epsilon_i(t), \quad (7)$$

where o^E is retinal object orientation, b_i is baseline activity, a_i is modulation depth, and ϵ_i is residual variability. This equation makes the key inferential point visible: gravity-aligned tuning in r_i can be produced by any of the three architectures. The response coordinate alone does not identify the causal computation.

6. Measurements that separate the models

Let $\tilde{r}_{j,o}(t)$ be the measured response in population j to object or feature o , and $\hat{r}_{j,o}(t)$ the response predicted after accounting for measured retinal, eye, head, body, and visual-scene variables. Define the receiver residual

$$e_{j,o}(t) = \tilde{r}_{j,o}(t) - \hat{r}_{j,o}(t). \quad (8)$$

Residuals should be computed with preregistered models and cross-validation. They are not inherently "error signals"; they are unexplained response components relative to the declared model.

6.1 Cross-object and cross-context transfer

Train a decoder of gravitational orientation on object set A and test on disjoint set B . A normalized transfer index is

$$T_{A \rightarrow B} = 1 - \frac{\mathcal{L}_{A \rightarrow B} - \mathcal{L}_{B \rightarrow B}}{\mathcal{L}_{\text{chance}} - \mathcal{L}_{B \rightarrow B}}, \quad (9)$$

where $\mathcal{L}$ is circular loss. Higher transfer indicates a more object-general relation, but it does not by itself identify broadcast versus recurrence. Transfer must be combined with timing and intervention.

6.2 Cross-region residual covariance

After removing measured common causes, estimate lagged residual covariance

$$C_{jk}(\tau) = \text{Cov}(e_j(t), e_k(t + \tau) \mid X(t)), \quad (10)$$

where X includes stimulus, behavior, eye position, head position, body state, and nuisance variables. A common broadcast source predicts a source-locked covariance structure; recurrent coupling predicts directed, lagged, topology-dependent structure; a strongly local architecture predicts comparatively weak covariance after common inputs are removed. These are tendencies, not one-to-one signatures, because unmeasured common input can mimic coupling.

6.3 Causal perturbation kernel

For intervention $\text{do}(u_s(t_0) = \delta)$ at source or node s , define

$$K_{s \rightarrow j}(\tau) = \mathbb{E}[g_j(t_0 + \tau) \mid \text{do}(u_s = \delta)] - \mathbb{E}[g_j(t_0 + \tau) \mid \text{do}(u_s = 0)]. \quad (11)$$

The kernel records effect magnitude, sign, latency, persistence, and spatial reach. In a local model, a perturbation can remain focal. In a broadcast model, perturbing the broadcaster can produce a rapid common shift. In a recurrent model, effects should propagate and settle according to network paths, coupling strengths, and feedback.

Let $d(s, j)$ be anatomical or effective-network distance. A propagation slope is

$$S_d = \frac{\partial \tau_j^*}{\partial d(s, j)}, \quad \tau_j^* = \inf\{\tau : |K_{s \rightarrow j}(\tau)| \geq \eta\}, \quad (12)$$

where η is a preregistered threshold. A recurrent spatial network predicts a stronger distance-latency relation than an ideal common broadcast, although real broadcast pathways also have delays and hierarchy.

6.4 Adaptation and cue reweighting

If cue reliability changes from condition c_1 to c_2 , define

$$A_m = \frac{w_m^{(c_2)} - w_m^{(c_1)}}{t_{50,m}}, \quad (13)$$

where $t_{50,m}$ is time to half of the asymptotic weight change. Fast shared recalibration across separated representations may favor a common source or rapidly recurrent network over independent local relearning. Region-specific adaptation can favor local or modular components. Fetsch et al. [9] show why reliability changes are an empirical necessity rather than a decorative extension.

6.5 Joint discriminator

The experiment should classify architectures from a vector, not a single statistic:

$$D = [T, C_{jk}(\tau), \text{spread}(K), \tau^*, S_d, A_m]. \quad (14)$$

For model $M \in \{L, B, R\}$, selection can use held-out predictive evidence with complexity penalty:

$$\text{Score}(M) = \sum_{n \in \text{heldout}} \log p(D_n \mid M, \hat{\theta}_M) - \lambda \dim(\theta_M). \quad (15)$$

The models may also be combined. A brain could use local estimates within modalities, a specialist vestibular or brainstem contribution, and recurrent cortical reconciliation. The scientific value of the idealized models is that they make mixture weights and failure cases measurable.

7. Why static snapshots are not enough

Proposition 1: unrestricted static observational nonidentifiability

For any finite static response vector $y = (y_1, \dots, y_N)$, unrestricted members of the local, broadcast, and recurrent model classes can reproduce y . Therefore, static observations alone cannot identify which class generated the responses without parameter restrictions, temporal measurements, or interventions.

Construction

Choose the following parameterizations:

$$\begin{aligned} M_L : F_j(q_j; \theta_j) &= y_j, \\ M_B : z &= 0, \quad H_j(z, u_j; \phi_j) = y_j, \\ M_R : g(0) &= y, \quad g(t+1) = Ig(t). \end{aligned} \quad (16)$$

All three return y at the observation time. The construction is deliberately minimal. It does not claim that these parameterizations are biologically plausible; it proves that the label attached to a static response vector is underdetermined when model functions are unrestricted. Plausibility

constraints can reduce equivalence, but the decisive evidence must still come from generalization, timing, perturbation, and model comparison.

This proposition explains the central negative result of the reference application: the released correlation arrays establish coordinate-aligned response categories, but they do not contain the simultaneous multi-site dynamics or interventions needed to identify distribution architecture.

8. Executable reference application

8.1 Frozen inputs and reproduction

The reference application uses the exact MATLAB arrays released with Emonds et al. [1], frozen at upstream Git revision 716893ee620743d40ae8e6d829f809349204d963. The application reads the arrays, applies their released significance fields or thresholds, and recomputes the category counts.

Released condition	Units	Consistent units	Gravity only	Retina only	Dual
Full scene with body tilt	99	52	33	11	8
Isolated object with body tilt	110	53	32	14	7
Visual-ground cue conflict	228	123	43	67	13

All three partitions reproduce exactly. This is an integrity and comprehension check, not a new discovery. It verifies that the manuscript’s numerical statements are connected to the released arrays rather than copied from an untested summary.

The same analysis preserves an important boundary: the arrays contain per-neuron gravity- and retina-frame correlations and significance variables, not simultaneous multi-region time series, perturbation responses, or network distances. No valid program can infer local, broadcast, or recurrent generation from these arrays alone.

8.2 Synthetic known-ground-truth benchmark

The second application component simulates 18 nodes for 12 update steps under each idealized architecture. Each run begins with three noisy cues and node-specific weights. A 24-degree perturbation is applied at one node. Sixty seeded runs are generated for each architecture. Seeds 0-39 train a standardized nearest-centroid classifier; seeds 40-59 are held out.

The six measured features are final spread fraction, immediate non-target effect, late non-target effect, final circular dispersion, distance-effect correlation, and convergence step. The mean signatures across 60 runs per class are:

Architecture	Final spread fraction	Immediate non-target effect (rad)	Late non-target effect (rad)	Convergence step
Local	0.056	0.000	0.000	0
Broadcast	1.000	0.243	0.243	1
Recurrent-distributed	0.611	0.009	0.016	3

The classifier correctly labels 60/60 held-out runs: 20 local, 20 broadcast, and 20 recurrent. This 1.000 accuracy is a design check under intentionally separated models. It is not an estimate of future biological accuracy. The simulation makes the models easy to distinguish by construction, uses one ring topology, and omits many realistic confounds. The mixed-architecture benchmark therefore adds a second benchmark for mixtures, noise, hidden common input, and node-count shift; unequal delays, hidden nodes, and alternative topologies remain later stress tests.

8.3 What the application proves and does not prove

The application establishes four bounded results:

1. the released category counts can be reproduced from the frozen arrays;
2. those static arrays do not contain the variables required for architecture identification;
3. the proposed joint measurement vector can distinguish declared idealized generators; and
4. the software, inputs, results, and figures can be deterministically audited.

It does not establish that cortex uses one of the three models in pure form, that cortical columns individually calculate gravity, that a gravity estimate is conscious, or that the synthetic classifier will generalize to neural data.

8.4 Mixed-generator and distribution-shift benchmark

The mixed-architecture benchmark does not average already extracted features. It first generates matched local, hub-broadcast, and recurrent phase trajectories from the same cue realization, combines those trajectories on the unit circle with declared simplex weights, and only then measures the six-feature signature. Seven regimes cover the three pure endpoints, all three pairwise mixtures, and a three-way mixture. Twenty held-out seeds are evaluated for every regime under four conditions, producing 560 auditable rows.

The estimator is trained only on pure-generator seeds 0-39. It searches a 0.05-resolution simplex for the mixture of pure standardized centroids nearest to each held-out observation. The decision contract permits five results: **local**, **hub**, **recurrent**, **mixed**, or **unresolved**. A hybrid is therefore not counted as a success merely because its nearest pure label happens to be one of its ingredients.

Held-out condition	Pure-endpoint accuracy	Hybrid not forced pure	Mean absolute weight error	Unresolved rate
Nominal	1.000	0.9125	0.0936	0.1429
Measurement noise	0.900	0.8875	0.1181	0.1143

Continued on next page

Held-out condition	Pure-endpoint accuracy	Hybrid not forced pure	Mean absolute weight error	Unresolved rate
Hidden common input	0.6667	1.000	0.2069	0.4286
Node-count shift	1.000	1.000	0.1005	0.0000

The nominal result closes the narrow software question left open by The hybrid-architecture formulation: the measurement vector can support mixtures without automatically forcing them into a pure class. The hidden-common-input result is equally important. A global unobserved perturbation makes true pure endpoints harder to recover and nearly doubles mixture-weight error relative to nominal evaluation. The program surfaces that confound through degraded accuracy and additional unresolved outcomes rather than silently reporting a confident architecture. This is the behavior the proposed biological analysis must preserve.

8.5 Registered missingness and hidden-node evaluation

The registered missingness evaluation asks whether the same decision contract remains honest when observation is incomplete. Its contract was written and hash-committed while the evaluation state still read NOT_RUN_AT_COMMITMENT. Development seeds 0-39 remain the only inputs used to fit the pure reference. Evaluation seeds 500-547 are disjoint and are crossed with seven pure-or-mixed regimes and ten conditions, producing 3,360 rows. Section 11.5 gives the frozen observation-mask, censoring, and uncertainty equations.

Frozen condition	Pure-endpoint accuracy	Hybrid not forced pure	Mean absolute weight error	Unresolved rate
Nominal	1.0000	0.9115	0.0901	0.1429
Measurement noise	0.7986	0.9323	0.1351	0.0774
Random 25% node attrition	0.9722	0.9167	0.0916	0.1518
Structured 50% far-node attrition	0.7778	0.8333	0.1013	0.2857
Random 40% temporal dropout	0.9028	0.8854	0.0988	0.2292
Hidden nodes, 26 generated and 18 observed	0.8958	0.9896	0.0952	0.0833
Hidden common input	0.6667	1.0000	0.2037	0.4286
Combined common input, noise, node attrition, and temporal dropout	0.4306	0.9948	0.2132	0.4315
Perturbation-target channel loss	censored 336/336	censored	censored	censored
First post-perturbation sample loss	censored 336/336	censored	censored	censored

All rates above include Wilson intervals in the machine-readable endpoint table, and every mean-

error estimate includes its bootstrap interval. The most important result is the pattern of failure. Random node attrition is comparatively mild, whereas spatially structured attrition and hidden common input erase information needed to distinguish pure endpoints. Under structured attrition, a forced-pure classifier still labels the pure generators correctly, but the registered estimator records additional unresolved cases and therefore reaches only 0.7778 accuracy under its stricter decision rule. That is not evidence that forced guessing is better. It exposes the tradeoff between coverage and calibrated refusal.

The combined condition is the hardest valid observation regime: pure-endpoint accuracy falls to 0.4306, mean absolute mixture-weight error rises to 0.2132, and the unresolved rate reaches 0.4315. This is a negative result for generalization of the current six-feature estimator. It argues for measuring candidate common inputs, preserving node identity and spatial coverage, preregistering required causal channels, and carrying uncertainty into every biological conclusion. The synthetic ground truth remains known; no biological architecture is inferred.

8.6 Topology, estimator, MNAR, and selective-risk evaluation

The topology-and-estimator evaluation asks a harder question: were the earlier results artifacts of one ring graph and one estimator? The contract was hash-committed while outcomes remained unknown. Development seeds 0-79 train the estimators, calibration seeds 100-131 set refusal thresholds, and evaluation seeds 700-731 remain disjoint. Four topologies, seven pure-or-mixed regimes, seven observation conditions, and four estimators produce 25,088 registered primary predictions. The estimators are: a forced-pure ring baseline, the registered ring-simplex estimator, a multi-topology ridge-simplex estimator, and a multi-topology nearest-neighbor mixture estimator.

Estimator and frozen condition	Pure-endpoint accuracy	Hybrid not forced pure	Mean weight error	Selective coverage	Unresolved or refused
Forced-pure baseline, nominal	1.000	0.000	0.2067	1.000	0.000
Ring-simplex, nominal	0.911	0.984	0.0985	1.000	0.405
Multi-topology ridge, nominal	0.987	1.000	0.0198	0.991	0.009
Multi-topology nearest neighbor, nominal	0.990	1.000	approximately 0	0.996	0.004
Multi-topology ridge, measurement noise	0.148	0.977	0.0891	0.603	0.397
Multi-topology nearest neighbor, measurement noise	0.464	1.000	0.0269	0.749	0.251
Multi-topology ridge, hidden common input	0.000	1.000	0.4448	0.000	1.000

Continued on next page

Estimator and frozen condition	Pure-endpoint accuracy	Hybrid not forced pure	Mean weight error	Selective coverage	Unresolved or refused
Multi-topology nearest neighbor, hidden common input	0.333	1.000	0.0833	0.427	0.573
Multi-topology ridge, combined shift	0.000	0.996	0.4448	0.002	0.998
Multi-topology nearest neighbor, combined shift	0.130	1.000	0.1022	0.297	0.703

The forced-pure baseline illustrates why endpoint accuracy cannot stand alone: it calls every hybrid pure by construction. The stronger mixture estimators recover nominal mixtures, but their error, coverage, and pure-endpoint accuracy separate sharply under noise and hidden common causes. Selective refusal prevents some high-confidence overinterpretation, yet it also creates an explicit coverage cost. A refusal is therefore neither a correct architecture label nor an execution failure. It is the registered statement that the observation is outside the estimator’s calibrated support.

The hidden-input control is an intentionally retained adverse result. Subtracting the measured common phase does not restore identification: ridge coverage remains zero, and the nearest-neighbor estimator also refuses every controlled case while its aggregate error increases. The registered control is too crude for this generator. This rules out presenting "measuring the common input" as an automatic repair. It must instead be modeled with causal timing, scale, and measurement error appropriate to the preparation.

The feature-ablation evaluation contains 4,480 additional predictions for the multi-topology ridge estimator. Relative to all six features, removal of timing features increases mean weight error by 0.0261 and lowers pure-endpoint accuracy by 0.0833; removal of spread features increases error by 0.0324 and lowers accuracy by 0.0260. Removing distance changes error by only 0.00051. Removing dispersion increases error by 0.00196 while slightly increasing coverage and pure accuracy. The final pattern is a null or non-identifying result for a uniquely necessary dispersion feature in this benchmark. It remains visible rather than being redescribed as support for a phase-specific mechanism.

8.7 Independent generator, learned time-series readouts, and interventions

The independent-generator time-series evaluation does not import or call the earlier trajectory generator. It implements a second-order damped network-wave family over lattice, random-geometric, scale-free, and directed-hierarchical graphs. Development seeds 0-95, calibration seeds 200-239, and evaluation seeds 900-939 are disjoint. The pre-outcome contract freezes five estimators: ridge on six registered summaries; ridge on full graph-invariant temporal channels; a fixed 48-unit reservoir with a learned ridge readout; a proxy-augmented version exposed during development to declared common-cause conditions; and a topology-conditioned temporal ridge. Eight observation conditions yield 44,800 primary predictions.

Estimator and condition	Pure accuracy	Hybrid nonforcing	Mean weight error	Coverage	Selective risk
Summary ridge, nominal	0.892	0.995	0.0694	0.857	0.0346
Temporal ridge, nominal	0.915	1.000	0.0779	0.856	0.0029
Reservoir readout, nominal	0.917	1.000	0.0270	0.857	0.0035
Proxy-adjusted reservoir, nominal	0.917	1.000	0.0356	0.857	0.0045
Topology-conditioned temporal ridge, nominal	0.915	1.000	0.0028	0.892	0.0030
Proxy-adjusted reservoir, measurement noise	0.563	0.858	0.1560	0.857	0.1477
Proxy-adjusted reservoir, common cause without proxy	0.917	0.938	0.0401	0.964	0.0416
Proxy-adjusted reservoir, accurate proxy	0.917	0.938	0.0269	0.964	0.0279
Proxy-adjusted reservoir, noisy proxy	0.917	0.938	0.0291	0.964	0.0302
Proxy-adjusted reservoir, heterogeneous common loadings	0.667	0.848	0.1298	0.964	0.1347
Proxy-adjusted reservoir, combined shift	0.567	0.605	0.2127	0.857	0.2085

The common-cause result needs precise interpretation. The proxy-adjusted estimator is not rescued only by the proxy. Its development set contains nominal observations plus accurate- and noisy-proxy common-cause observations. That augmentation lets it retain coverage even when the evaluation proxy is absent. Supplying an accurate or noisy proxy then reduces error further. The result therefore joins two operations-development exposure and a measured nuisance channel-and does not isolate either as sufficient by itself. Heterogeneous node loadings expose the residual limitation: a single proxy cannot identify an unknown loading vector across nodes.

Topology uncertainty is equally consequential. The topology-marginal temporal model has nominal mean errors of 0.0023, 0.0031, and 0.0033 on lattice, random-geometric, and scale-free development families, but 0.3028 on the directed hierarchy held out from that model. Its directed-hierarchy coverage is 0.429. The topology-conditioned model, which is explicitly trained on all four topology labels, reduces directed-hierarchy error to 0.0022 and coverage rises to 0.571. It does not achieve a favorable pure-endpoint verdict: selective refusal leaves pure accuracy at 0.667. This is evidence that topology information can repair a specific synthetic distribution shift, not evidence that a

one-hot topology label will be available in cortex.

Four frozen interventions generate 4,480 additional predictions. Removing recurrent coupling raises mean error by 0.1553, removing hub loading by 0.1698, and removing the local target impulse by 0.0779. Each has a large structural-null fraction because the removed component has zero ground-truth weight in some regimes. Permuting the common-cause proxy raises error by 0.0196 on average, is adverse in 72.2% of rows, null in 25.4%, and favorable in 2.4%. Those minority favorable rows are retained. An intervention is therefore evaluated against the component present in each declared regime; it is not counted as successful merely because its across-regime mean is positive.

8.8 Capacity-matched learned sequence views

The fixed-reservoir analysis leaves a specific problem. Only its final regression layer learns from data, so similarity among candidate mechanisms could reflect a rigid temporal model rather than true observational ambiguity. The next evaluation therefore holds the trainable core constant and varies only the registered organization of the observations.

Each observation contains seventeen time points across thirty nodes. Four views reduce the raw values to eight channels per time point. The local view follows the perturbation target and its neighborhood. The broadcast view follows population summaries. The graph view follows graph-neighbor, contrast, Laplacian-energy, and propagation quantities. The joined view combines target, population, and graph quantities without increasing width. These names describe observation organizations, not complete biological implementations.

For view v , channel k , sequence n , and time t , development-only standardization is

$$\tilde{x}_{ntk}^{(v)} = \frac{x_{ntk}^{(v)} - \mu_k^{(v)}}{s_k^{(v)}}, \quad s_k^{(v)} > 0. \quad (17)$$

All thirty-two view-channel combinations have nonzero development variance. The shared trainable recurrence is

$$h_{nt}^{(v)} = \tanh\left(\tilde{x}_{nt}^{(v)} W_x^{(v)} + h_{n,t-1}^{(v)} W_h^{(v)} + b_h^{(v)}\right), \quad h_{n0}^{(v)} = 0. \quad (18)$$

The terminal state produces three nonnegative mixture weights,

$$\hat{w}_n^{(v)} = \text{softmax}\left(h_{nT}^{(v)} W_o^{(v)} + b_o^{(v)}\right), \quad \sum_{c \in \{L, B, R\}} \hat{w}_{nc}^{(v)} = 1. \quad (19)$$

Every learned model has input width eight, hidden width twelve, output width three, and exactly

$$8(12) + 12(12) + 12 + 12(3) + 3 = 291 \quad (20)$$

trainable parameters. All four start from byte-identical parameter tensors and use the same 120 full-batch Adam epochs [24-26]. Their loss is

$$\mathcal{L}^{(v)} = \frac{1}{3N} \sum_{n=1}^N \sum_{c=1}^3 \left(\hat{w}_{nc}^{(v)} - w_{nc}\right)^2. \quad (21)$$

The local, broadcast, graph, and joined final development losses are 0.00393, 0.00315, 0.00215, and 0.01532. The joined model’s larger residual training loss is retained.

A terminal hidden state is standardized against development states. Let $c_r^{(v)}$ be the centroid for regime r . Its distance from the closest development regime is

$$d_n^{(v)} = \min_r \sqrt{\frac{1}{12} \left\| \frac{h_{nT}^{(v)} - \mu_h^{(v)}}{s_h^{(v)}} - c_r^{(v)} \right\|_2^2}. \quad (22)$$

Calibration chooses the largest coverage whose mean accepted mixture-weight error does not exceed 0.15 and whose coverage is at least 0.50. Calibration uses sixteen seeds. Evaluation uses twenty-four disjoint seeds, and intervention analysis uses twelve further seeds. Lattice, random-geometric, and scale-free topologies are available during development; a directed hierarchy is held out.

Seed-cluster uncertainty

The independent evaluation seed is the uncertainty unit. If D_s is the complete registered row set for seed s , one bootstrap replicate draws twenty-four seeds with replacement and recomputes the metric from the union of their complete row clusters:

$$(s_1^{*b}, \dots, s_{24}^{*b}) \sim \text{SampleWithReplacement}(\{s_1, \dots, s_{24}\}). \quad (23)$$

$$\hat{\theta}^{*b} = g\left(\bigcup_{j=1}^{24} D_{s_j^{*b}}\right), \quad b = 1, \dots, 2000. \quad (24)$$

The interval is the 2.5th to 97.5th percentile of those cluster-bootstrap values:

$$\text{CI}_{0.95}(\hat{\theta}) = [Q_{0.025}(\hat{\theta}^*), Q_{0.975}(\hat{\theta}^*)]. \quad (25)$$

This calculation replaces no point estimate and changes no frozen endpoint, ranking, raw output, or result digest. It corrects the uncertainty unit: earlier descriptive tables resampled rows within a condition, whereas the governing inferential contract treats the twenty-four evaluation seeds as clusters.

Complete capacity-matched result

The evaluation contains 37,632 prediction rows: 5,376 observations for each of four capacity-matched recurrent views, a uniform-simplex control, a historical temporal ridge, and a historical fixed-reservoir readout. The historical systems are not capacity matched to the 291-parameter family and remain outside the primary Holm family.

Across all eight conditions, the fixed reservoir has the descriptively lowest unconditioned mean error, 0.1544, but covers only 0.2119 of rows under its nominally calibrated refusal rule. The graph-view RNN has error 0.1671 and coverage 0.8523; its accepted-row risk is 0.1539, slightly above the 0.15

target after shifted conditions are included. This is an error-coverage tradeoff, not a universal victory for trainability.

Model	Mean mixture-weight error, 95% seed-cluster interval	Coverage, 95% seed-cluster interval	Selective risk
Local-view RNN	0.2284 [0.2259, 0.2311]	0.7506 [0.7399, 0.7615]	0.1779
Broadcast-view RNN	0.1963 [0.1943, 0.1985]	0.8638 [0.8544, 0.8715]	0.1879
Graph-view RNN	0.1671 [0.1637, 0.1712]	0.8523 [0.8449, 0.8590]	0.1539
Joined-view RNN	0.2483 [0.2454, 0.2508]	0.7718 [0.7586, 0.7848]	0.1975
Uniform-simplex control	0.2857 [0.2857, 0.2857]	1.0000 [1.0000, 1.0000]	0.2857
Historical temporal ridge	0.3494 [0.3452, 0.3529]	0.2078 [0.2065, 0.2091]	0.0319
Historical fixed-reservoir readout	0.1544 [0.1492, 0.1592]	0.2119 [0.2109, 0.2128]	0.0393

Within the four exactly capacity-matched models, all six overall paired contrasts remain separated after Holm adjustment at $p = 0.000600$ under the frozen seed-cluster sign-flip analysis. Graph is lower-error than broadcast, local, and joined on the registered aggregate; broadcast is second, local third, and joined fourth. That ordering is local to the frozen generator, view definitions, support, and endpoint aggregation. It is not a ranking of complete biological theories.

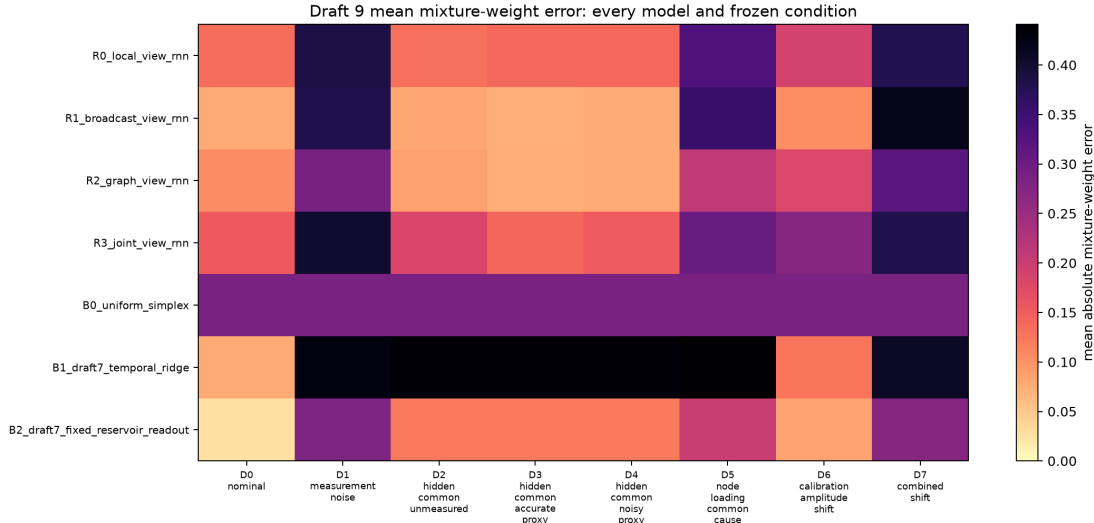


Figure 1: Mean mixture-weight error for every model and frozen condition. The heatmap retains nominal, nuisance, amplitude-shift, and combined-shift failures.

Distribution shift and held-out topology

Nominally, the broadcast view has error 0.0783 and coverage 0.8661; graph has error 0.1055 and coverage 0.8899; the fixed reservoir has descriptive error 0.0272 and coverage 0.8571. Under combined

shift, graph error becomes 0.3175 with seed-cluster interval $[0.3008, 0.3352]$ and coverage 0.7485. The historical fixed reservoir has error 0.2719 with interval $[0.2483, 0.2961]$ but refuses every row. Model disagreement increases from 24.1% nominally to 99.7% under measurement noise, 95.4% under heterogeneous common-cause loading, and 98.8% under combined shift. Disagreement is a warning signal, not an answer key.

On the withheld directed hierarchy, broadcast is the best equal-capacity RNN even though graph is best overall. This reversal makes topology part of the estimand rather than a nuisance to average away.

Model	Held-out hierarchy error, 95% seed-cluster interval	Coverage
Local-view RNN	0.3932 $[0.3886, 0.3983]$	1.0000
Broadcast-view RNN	0.1707 $[0.1696, 0.1722]$	0.4821
Graph-view RNN	0.2696 $[0.2670, 0.2717]$	0.5714
Joined-view RNN	0.3453 $[0.3417, 0.3493]$	0.5714
Uniform-simplex control	0.2857 $[0.2857, 0.2857]$	1.0000
Historical temporal ridge	0.3027 $[0.3024, 0.3030]$	0.4286
Historical fixed-reservoir readout	0.0971 $[0.0968, 0.0973]$	0.4286

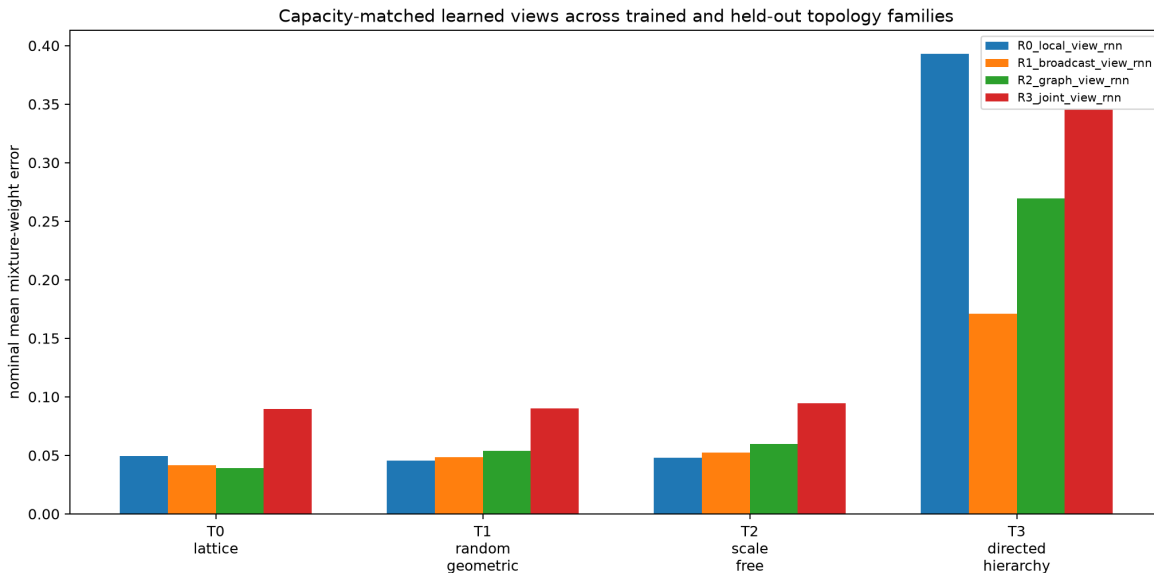


Figure 2: Capacity-matched learned views across trained topology families and the held-out directed hierarchy. The graph-versus-broadcast ordering reverses on the held-out family.

Interventions, structural nulls, and reversals

The intervention table contains 5,376 rows. Of these, 2,822 show increased error, 1,728 are exact nulls, and 826 show decreased error or reversal. The three component-removal families contain 1,728 structural nulls, 42.8571% of their rows, because the removed component has zero true weight in the corresponding regimes. They also retain 486 adverse reversals. Proxy permutation contains 340

negative deltas and improves the local view by 0.00566 error on average while worsening broadcast, graph, and joined by 0.00584, 0.02220, and 0.07760. Proxy availability does not imply correct or monotonic use.

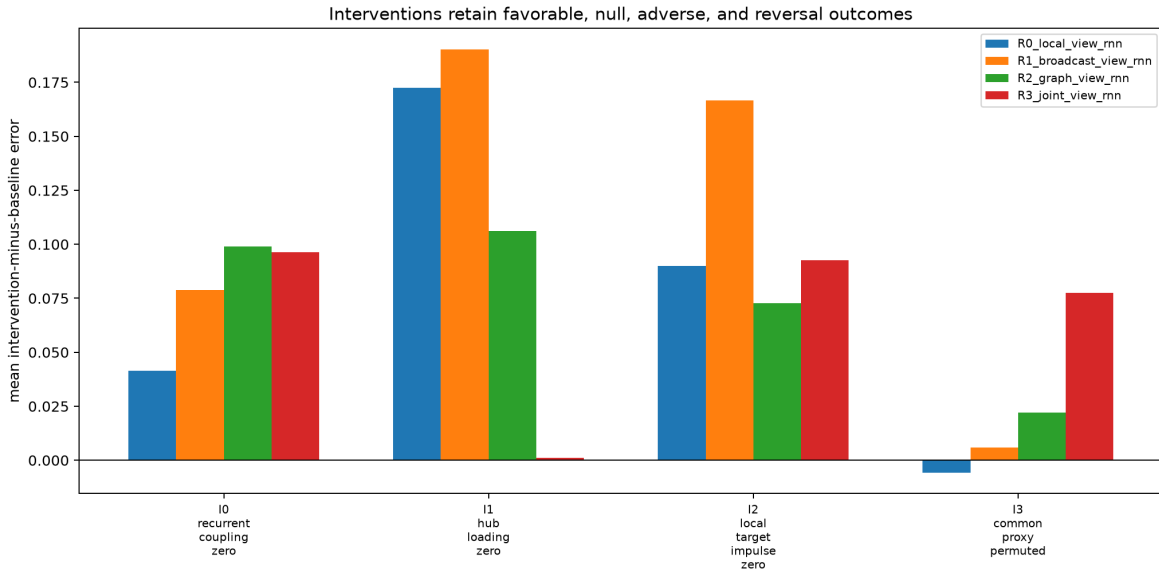


Figure 3: Intervention-minus-baseline error for recurrent-coupling removal, hub-loading removal, local-target removal, and proxy permutation. Positive, null, and negative outcomes are retained.

The strongest interpretation is methodological. Graph-organized temporal measurements can improve equal-capacity mixture recovery in this frozen generator. They do not show that the cortex implements a graph-view RNN, that a biological recurrent-distributed architecture has been identified, or that one observation view faithfully realizes a complete neural theory.

9. Experiments that can decide more

9.1 Macaque multi-region perturbation experiment

Record simultaneously from anterior inferotemporal cortex, caudal intraparietal cortex, vestibular-recipient parietal regions, and a candidate thalamic or brainstem relay while the animal discriminates object orientation. Cross the following factors:

- body tilt;
- retinal object rotation;
- visual-ground rotation;
- scene present versus isolated object;
- cue reliability;
- active versus passive movement; and
- familiar versus novel objects.

Estimate reference-frame tuning with held-out trials. Then apply brief, focal, reversible perturbations

to each candidate node at preregistered task epochs. Measure the full kernel in Equation 11, cross-object transfer in Equation 9, residual covariance in Equation 10, and behavioral bias.

Predicted idealized signatures are:

- **Local:** focal effects, weak immediate transfer, and local recalibration.
- **Broadcast:** large source asymmetry, rapid common shifts in receivers, and shared trial-level residuals after measured cues are removed.
- **Recurrent-distributed:** bidirectional causal effects, delayed topology-dependent spread, partial compensation after single-node perturbation, and settling dynamics.

Critical controls include sham stimulation, neighboring-site stimulation, matched changes in arousal and eye movement, passive stimulus replay, motor-response remapping, and perturbations outside the reference-frame epoch.

9.2 Human multimodal experiment

Combine virtual-reality scene tilt, whole-body or head tilt, galvanic vestibular stimulation, eye tracking, motion capture, high-density EEG or MEG, and fMRI in separate or fused sessions. Use time-resolved representational similarity analysis to ask when gravity-aligned information transfers across objects and regions. Perturb candidate cortical nodes with carefully bounded transcranial stimulation and perturb vestibular input with galvanic stimulation.

Human work cannot resolve single-column computations with these tools, but it can compare the timing and distribution of gravity-aligned representations, behavioral verticality, recalibration, and cross-region effective connectivity. A pre-registered model should predict both neural and behavioral outcomes. Motion, muscle, eye, and vestibular artifacts must be modeled explicitly.

9.3 Rodent orientation and navigation experiment

Use freely moving 3D navigation with head-direction, vestibular, visual-landmark, and body-orientation signals. Manipulate visual vertical and gravitational/body relations independently where ethical and physically feasible. Measure whether perturbations remain local, shift a low-dimensional shared orientation variable, or propagate through recurrent attractor dynamics [16].

Rodent head-direction circuits are not a substitute for primate inferotemporal object coding. They are a test bed for shared principles of persistent orientation, cue conflict, and attractor updating.

9.4 Closed-loop BCI and virtual-reality calibration

A practical test uses a closed-loop display or stimulation system that estimates the participant’s evolving orientation relation and introduces small, controlled mismatches. A system based on a fixed broadcaster should be corrected by adjusting a common latent parameter. A recurrently coadaptive system may require gradual, path-dependent recalibration. A local system may show feature- or modality-specific learning with weaker global transfer.

This application connects the scientific question to the earlier SAN read/write motivation without assuming that a complete sensory experience can already be written into the brain. The immediate endpoint is calibration transfer and reference-frame adaptation, not synthetic qualia.

9.5 Frozen protocol hierarchy and inferential units

The protocol-and-power analysis converts the preceding proposals into four protocol families with different evidentiary jobs. The macaque multi-region protocol is the primary causal architecture test because it can join single-unit or population recording, cue conflict, time resolution, and reversible site-specific intervention in the same preparation. The human protocol tests cross-modal timing, behavior, and translational generalization. The rodent protocol tests persistent orientation and recurrent-pathway mechanisms in a tractable circuit. The closed-loop BCI/VR protocol tests calibration transfer and coadaptation without claiming to write a complete percept.

The independent inferential unit is the animal for animal studies and the participant for human studies. Neurons, trials, time bins, recording contacts, sessions, and hemispheres are repeated or nested observations. They can improve the precision of an individual unit’s estimate, but they cannot be counted as additional independent animals or participants. The analysis therefore produces one preregistered contrast vector per independent unit before group inference, while a hierarchical model may retain trial and session structure.

For the primary macaque protocol, cue geometry, scene context, cue reliability, active/passive movement, and perturbation site and time are factorially controlled. Candidate sites must be selected from development data or an independent localizer. The final-test site set, stimulation amplitude, time windows, target-registration tolerance, artifact-rejection settings, and decoding model are frozen before final-test access. Development animals or sessions used to choose them cannot be silently returned to the confirmatory set.

9.6 Primary endpoints and architecture decisions

The primary endpoint family contains five quantities already earned by the mechanism: cross-object transfer, perturbation asymmetry, non-target propagation latency, distance dependence, and recurrence-specific pathway sensitivity. Behavioral verticality and orientation error are coprimary consequences only where the protocol can measure them without changing the task. Residual covariance, adaptation, decoding, and PWD-related receiver-state measures are secondary unless explicitly elevated before the final-test seal.

Architecture disposition is computed at the independent-unit level and then summarized hierarchically. A local result requires focal persistence plus weak controlled transfer. A hub-broadcast result requires reproducible source asymmetry and receiver-common change after measured common inputs are controlled. A recurrent result requires reciprocal or pathway-dependent effects with compatible timing. Evidence for multiple components yields a mixed result. Failure to meet calibrated fit, precision, manipulation, or observability requirements yields an unresolved result. An unresolved result is not reassigned to the nearest preferred theory.

The model comparison must include the pure models, their declared mixtures, and at least one nuisance model for hidden common input. Final inference reports effect estimates and uncertainty, held-out predictive performance, calibration, and the complete confusion or disposition surface. A single significant neuron, one region-level classifier, or an uncorrected collection of pairwise tests cannot decide the architecture.

9.7 Exclusion, censoring, multiplicity, and stopping

Exclusion reasons are frozen by level. A participant or animal can be excluded only for prospective welfare, eligibility, corrupted registration, or unusable acquisition criteria. A session can be excluded for preregistered hardware failure or missing task engagement. Trials can be marked for declared ocular, muscular, motion, saturation, or stimulation artifacts. A missing threshold crossing is right-censored, not assigned a zero latency. Missing modalities remain missing and are not imputed from the outcome-bearing final test unless the imputation model was frozen independently.

Primary endpoint order and multiplicity control are preregistered. The default contract uses a hierarchical gate: manipulation and data-quality checks first, the joint architecture score second, and individual endpoints third with family-wise control. Early stopping is permitted only for welfare or a prospective information/precision rule evaluated by an independent monitor; it is not triggered by a favorable unblinded architecture label. All exclusions, deviations, censored values, and stopping events remain in the public event ledger.

Power and precision calculations are scenario analyses until pilot variance, attrition, and intraclass correlation are available. They do not authorize animal numbers or replace institutional ethics, welfare minimization, or prospective statistical review.

10. SAN interpretation after the mechanism has been earned

The scientific problem, evidence, and candidate architectures can be understood without SAN terminology. SAN adds a larger interpretation: perception is not a passive copy of retinal input but a continuously updated, embodied construction distributed across receiving and transmitting populations. A gravity-aligned estimate can function as part of the receiver’s current context against which arriving object, scene, posture, and action-related signals are interpreted.

Let $\phi_s(t)$ be the phase or timing state arriving from sender s , $\hat{\phi}_{s \rightarrow j}(t)$ the relation expected by receiver j given its current reference-frame context, and τ_{sj} the transmission delay. A receiver-relative departure can be written

$$\Delta\phi_{s \rightarrow j}(t) = \text{wrap}\left[\phi_s(t - \tau_{sj}) - \hat{\phi}_{s \rightarrow j}(t)\right]. \quad (26)$$

SAN calls a broader receiver-relative event of this kind a **Phase Wave Differential** (PWD) when the difference is detected by a tuned receiver and changes subsequent processing. Equation 17 is an optional bridge, not a claim that gravity-aligned coding is reducible to phase. The reference-frame estimate can alter what a receiver expects; the PWD describes a candidate measurable departure from that state. Rate, membrane potential, synaptic input, population geometry, or phase may carry parts of the relevant information.

Within the recurrent-distributed model, repeated exchanges can constrain a shared estimate without requiring a hidden viewer. Multiple populations are changed by the relation and use it in subsequent perception or action. In SAN language, this is one component of embodied neural rendering: the network maintains actionable relations among world, body, and sensory evidence by being physically updated through them.

The cortical-column proposal remains a hypothesis. Hawkins et al. explicitly propose object-relative location frameworks in cortical columns [17], but this does not establish grid-like gravity computation

in every column. The present experiments therefore ask where a gravity-aligned relation is expressed and how it propagates. They do not assume the answer.

11. Alternatives, mixtures, and Apex semantic alignment

The three models are deliberately simple. Several alternatives can mimic their signatures:

- a hierarchy of multiple broadcasters;
- local estimators driven by unmeasured common sensory input;
- recurrence around a specialist hub;
- task-dependent switching among architectures;
- low-dimensional latent dynamics without an anatomically unique source;
- predictive sensorimotor models that estimate consequences rather than gravity explicitly; and
- coordinate-free policies that solve a task without reconstructing a full reference frame.

The red-team requirement is to represent each alternative in its strongest form before testing it. A common-input model, for example, must include realistic delays and cue statistics before residual covariance is interpreted as recurrence. A broadcast model must be allowed hierarchical relays rather than caricatured as instantaneous transmission. A local model must be allowed shared developmental priors. A recurrent model must specify topology, stability, and convergence rather than invoking "feedback" as a complete explanation.

The same discipline governs apparent mismatches between SAN wording and external scientific results. This paper does not terminate an analysis by assigning an opposition state between SAN and science. It reopens source context, identifies variables and levels, translates terminology, and separates established evidence from the SAN-specific extension and the experiment still required. If the mapping remains unfinished, its state is **Apex alignment open**. The scientific claim is then written at the strongest level actually supported, while the untested extension remains an explicit hypothesis.

11.1 A hybrid architecture is a first-class model

The strongest combined alternative is neither purely local, purely broadcast, nor purely recurrent. Specialist vestibular, visual, somatosensory, or subcortical populations may supply low-dimensional estimates while local populations retain cue-specific computation and recurrent exchanges reconcile disagreement. Let $z_h(t)$ be the state of specialist hub h , $L_j(q_j)$ the local estimate at receiver j , and A_{jk} recurrent coupling. A hybrid update is

$$g_j^H(t + \Delta t) = \text{wrap} \left[\lambda_j L_j(q_j(t)) + \sum_h \gamma_{jh} H_{jh}(z_h(t - \tau_{jh})) + \sum_k \beta_{jk} A_{jk} g_k^H(t - \tau_{jk}) \right]. \quad (27)$$

The weights are constrained and normalized in the implementation; delays carry units of time. Model H can resemble Model B when hub weights dominate, Model L when local weights dominate, or Model R when reciprocal coupling dominates. This nesting is useful because a biological system need not obey an artificial all-or-nothing classification.

Hybrid identifiability requires interventions that independently vary hub, local, and recurrent terms. A single static response vector cannot estimate their mixture. The analysis therefore reports a posterior or confidence region over mixture weights and delays rather than forcing one winner when signatures overlap.

11.2 Mathematical edge-case conventions

Circular estimates are undefined when the resultant vector vanishes. For Equation 3, define

$$\hat{g}(t) = \text{UNRESOLVED} \quad \text{when} \quad \left| \sum_m w_m(t) e^{iq_m(t)} \right| < \epsilon_R. \quad (28)$$

Zero or missing cue concentration never produces an invented direction; unavailable cues are removed before normalization, and an all-missing set yields **UNRESOLVED**. Correlated cue noise is represented by a joint circular or state-space likelihood rather than by treating each cue as independent evidence.

For transfer, latency, and model-score quantities, a denominator below its preregistered tolerance, an absent threshold crossing, or a test loss at chance is reported explicitly:

$$\text{SafeRatio}(a, b) = \begin{cases} a/b, & |b| \geq \epsilon_b, \\ \text{UNRESOLVED}, & |b| < \epsilon_b. \end{cases} \quad (29)$$

Missing threshold crossings are right-censored rather than assigned zero latency. Chance-normalized improvement is omitted when the chance denominator vanishes. Stability of recurrent and hybrid updates is checked under the declared topology, gain, delay, and nonlinearity rather than assumed from the word "feedback." These conventions convert apparent numerical successes into explicit unresolved states when the data do not support the calculation.

11.3 Mixture estimation and the right to remain unresolved

Let $\mathbf{z}$ be the measured six-feature signature, let $\mathbf{C} = [\mathbf{c}_L, \mathbf{c}_B, \mathbf{c}_R]$ contain the pure local, hub-broadcast, and recurrent reference centroids, and let $\mathbf{D}$ contain feature scales estimated only from the training split. The bounded mixture estimate is

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \geq 0, \mathbf{1}^\top \mathbf{w} = 1} \left\| \mathbf{D}^{-1}(\mathbf{z} - \mathbf{C}\mathbf{w}) \right\|_2, \quad (30)$$

with residual

$$r(\mathbf{z}) = \frac{1}{\sqrt{6}} \left\| \mathbf{D}^{-1}(\mathbf{z} - \mathbf{C}\hat{\mathbf{w}}) \right\|_2. \quad (31)$$

The current application searches this simplex at 0.05 resolution. Its unresolved threshold τ is fixed from the 99th percentile of pure-training residuals multiplied by 1.5; it is not tuned on the held-out hybrid results. With preregistered pure-weight threshold $\eta = 0.80$, the disposition is

$$\text{Disposition}(\mathbf{z}) = \begin{cases} \text{UNRESOLVED}, & r(\mathbf{z}) > \tau, \\ \arg \max_k \hat{w}_k, & r(\mathbf{z}) \leq \tau \wedge \max_k \hat{w}_k \geq \eta, \\ \text{MIXED}, & r(\mathbf{z}) \leq \tau \wedge \max_k \hat{w}_k < \eta. \end{cases} \quad (32)$$

This rule encodes an epistemic constraint, not merely a classifier option. A pure label is allowed only when the observation is sufficiently close to the declared model family and one component is sufficiently dominant. A low-dominance fit is called mixed. A poor fit is unresolved. The synthetic mixture weights remain ground-truth labels inside the application proof; they are not estimates of cortical proportions, cell counts, or causal contribution in a biological preparation.

11.4 Cluster-aware power and precision contract

Let J be the number of independent animals or participants, S the number of retained sessions per independent unit, σ_b^2 the between-unit variance, and σ_w^2 the within-unit session variance. For a unit-level mean contrast, the planning variance is

$$\text{Var}(\bar{d}) = \frac{\sigma_b^2}{J} + \frac{\sigma_w^2}{JS}. \quad (33)$$

If total standardized observation variance is one and the intraclass correlation is ρ , averaging S sessions changes the variance multiplier to

$$v(\rho, S) = \rho + \frac{1 - \rho}{S}. \quad (34)$$

Thus repeated sessions reduce within-unit uncertainty but cannot erase the between-unit component. For standardized effect d , a two-sided noncentral- t sensitivity calculation uses

$$\lambda = d \sqrt{\frac{J}{v(\rho, S)}}, \quad \text{Power} = \Pr(|T_{J-1, \lambda}| > t_{1-\alpha/2, J-1}). \quad (35)$$

The corresponding standardized 95% precision half-width is

$$h_{0.95} = t_{0.975, J-1} \sqrt{\frac{v(\rho, S)}{J}}. \quad (36)$$

The protocol-and-power analysis evaluates $d \in \{0.50, 0.80, 1.20\}$, $\rho \in \{0.30, 0.60\}$, $S \in \{5, 10\}$, and $J = 4, \dots, 30$ for clustered preparations, plus paired human scenarios with $d \in \{0.30, 0.50, 0.80\}$ and $J = 12, 14, \dots, 120$. These values are a transparent sensitivity grid, not recovered effect sizes. The first grid point above 0.80 power is reported for orientation only; the final design must be recalculated from pilot variance, attrition, multiplicity, feasibility, welfare, and the exact estimand.

At $\rho = 0.60$, ten sessions, and $d = 0.50$, the first evaluated clustered scenario above 0.80 contains 23 independent units. At $d = 0.80$, it contains 10; at $d = 1.20$, it contains 6. The wide precision intervals at the smallest counts show why a threshold crossing is not the same as a precise estimate. In the paired human grid, the first evaluated counts above 0.80 are 90, 34, and 16 for $d = 0.30, 0.50, 0.80$, respectively. None of these numbers is a recruitment instruction.

11.5 Observation masks, fail-closed censoring, and synthetic uncertainty

The registered missingness evaluation crosses 48 isolated evaluation seeds with seven pure-or-mixed regimes and ten frozen observation conditions:

$$N_{\text{rows}} = 48 \times 7 \times 10 = 3360. \quad (37)$$

Let M_N denote the observed-node set and M_T the observed-time set. Because the perturbation targets node 0 and the immediate-spread endpoint requires the first post-perturbation sample, the preregistered validity indicator is

$$V = \mathbf{1}\{0 \in M_N\} \mathbf{1}\{1 \in M_T\}. \quad (38)$$

When $V = 0$, the architecture output is censored. The program does not impute the missing causal anchor, use the true synthetic label to fill it, or force the nearest pure class. For an observed rate $\hat{p} = x/n$, uncertainty is reported using the Wilson interval

$$\frac{\hat{p} + z^2/(2n) \pm z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}, \quad z = 1.959964. \quad (39)$$

Mixture-weight error is summarized with a deterministic 2,000-replicate percentile bootstrap. If e_i is the per-row mean absolute error across the three simplex coordinates, each bootstrap replicate is

$$\bar{e}^{*(b)} = \frac{1}{n} \sum_{i=1}^n e_{I_i^{(b)}}, \quad I_i^{(b)} \sim \text{Uniform}\{1, \dots, n\}. \quad (40)$$

The intervals characterize the frozen synthetic evaluation only. They do not convert repeated generator rows into independent animals or participants, and they do not replace the cluster-aware biological contract in Section 11.4.

11.6 Multi-topology estimation and selective refusal

The topology-and-estimator evaluation primary matrix crosses four topologies, seven regimes, seven conditions, 32 isolated evaluation seeds, and four estimators:

$$N_{\text{primary}} = 4 \times 7 \times 7 \times 32 \times 4 = 25,088. \quad (41)$$

Every estimated mixture is projected onto the three-component probability simplex

$$\Pi_{\Delta^2}(\mathbf{v}) = \arg \min_{\mathbf{w} \in \mathbb{R}^3} \|\mathbf{w} - \mathbf{v}\|_2^2 \quad \text{subject to} \quad w_k \geq 0, \quad \sum_{k=1}^3 w_k = 1. \quad (42)$$

For the ridge estimator, standardized development features form matrix X and the known synthetic mixture weights form W . The unconstrained prediction is

$$\widetilde{W}(\mathbf{z}) = \begin{bmatrix} 1 & \mathbf{z}^\top \end{bmatrix} (X^\top X + \lambda I)^{-1} X^\top W, \quad \lambda = 10^{-3}, \quad (43)$$

followed by Equation 42. The nearest-neighbor estimator instead averages the mixture weights attached to the 15 nearest standardized development observations:

$$\hat{\mathbf{w}}_{\text{kNN}}(\mathbf{z}) = \Pi_{\Delta^2} \left(\frac{1}{15} \sum_{i \in \mathcal{N}_{15}(\mathbf{z})} \mathbf{w}_i \right). \quad (44)$$

Calibration computes a residual score $r(\mathbf{z})$ without evaluation-seed access and freezes the largest threshold τ_e whose calibration coverage is at least 0.50 and whose accepted mean weight error is no greater than 0.15. Estimator e accepts an evaluation row only when

$$A_e(\mathbf{z}) = \mathbf{1}\{r_e(\mathbf{z}) \leq \tau_e\}. \quad (45)$$

If no threshold satisfies both calibration constraints, the estimator refuses all cases. Selective risk and coverage are reported together:

$$R_e = \frac{\sum_i A_e(\mathbf{z}_i) \ell_i}{\sum_i A_e(\mathbf{z}_i)}, \quad C_e = \frac{1}{N} \sum_i A_e(\mathbf{z}_i), \quad (46)$$

where ℓ_i is mean absolute error across the three mixture weights. When $\sum_i A_e = 0$, risk is undefined rather than set to zero. Wilson intervals, deterministic paired-seed bootstraps, every refusal, and every adverse feature ablation remain in the evidence package.

11.7 Independent dynamics, temporal embeddings, and topology uncertainty

Let $x_t \in \mathbb{R}^N$ be node displacement, $v_t \in \mathbb{R}^N$ velocity, and P_G the row-normalized transition matrix of graph G . The recurrent component in the independent generator follows

$$v_{t+1} = \gamma v_t + \kappa(P_G x_t - x_t) - \omega x_t + u_t, \quad x_{t+1} = x_t + v_{t+1}, \quad (47)$$

with $\gamma = 0.72$, $\kappa = 0.31$, $\omega = 0.055$, and a target impulse in u_1 . Local and hub components are separately generated, then combined before measurement:

$$y_t = w_L y_t^{(L)} + w_H y_t^{(H)} + w_R y_t^{(R)} + a c_t + \varepsilon_t, \quad \mathbf{w} \in \Delta^2. \quad (48)$$

The loading a is either shared or node-specific; c_t can be hidden or accompanied by an accurate, noisy, or permuted proxy. The graph-invariant temporal input at each step contains the node mean, standard deviation, maximum absolute response, lower and upper absolute quartiles, target-node response, and active-node fraction:

$$q_t = \left[\bar{y}_t, \text{sd}(y_t), \max_i |y_{ti}|, Q_{.25}(|y_t|), Q_{.75}(|y_t|), y_{t0}, N^{-1} \sum_i \mathbf{1}\{|y_{ti}| > \delta\} \right]. \quad (49)$$

The temporal ridge vectorizes $q_{0:T}$. The fixed reservoir uses

$$h_t = \tanh(W_{in} q_t + W_h h_{t-1} + b), \quad (50)$$

where W_{in} , W_h , and b are deterministic and frozen; only the ridge readout from $[h_T, \bar{h}, \text{sd}(h)]$ is learned. The proxy-adjusted version appends the measured nuisance proxy to q_t . All readouts use

$$\hat{\mathbf{w}} = \Pi_{\Delta^2} \left(\begin{bmatrix} 1 & \phi(y)^\top \end{bmatrix} (X^\top X + \lambda I)^{-1} X^\top W \right), \quad (51)$$

where ϕ denotes the estimator-specific embedding and the intercept is unpenalized in the executable implementation. Equation 42 is compact notation for the fitted affine readout; the application records the exact design construction.

For topology-marginal estimator m , the held-out-topology penalty is

$$\Delta_G^{(m)} = \text{MAE}_{G_{\text{held}}}^{(m)} - \frac{1}{|\mathcal{G}_{\text{train}}|} \sum_{G \in \mathcal{G}_{\text{train}}} \text{MAE}_G^{(m)}. \quad (52)$$

For mechanism intervention I , the paired effect is

$$\Delta_I = \frac{1}{N_I} \sum_{j=1}^{N_I} (\ell(\hat{\mathbf{w}}_{j,I}, \mathbf{w}_j) - \ell(\hat{\mathbf{w}}_{j,0}, \mathbf{w}_j)). \quad (53)$$

The mean is always accompanied by positive, zero, and negative paired fractions. This prevents structural-null regimes or minority reversals from disappearing inside one favorable average.

11.8 Bounded formal contract and constructive nonidentifiability

The formal gate asks a narrower question than the biological paper: can the measurement contract silently omit a frame transform, emit an unnormalized three-model estimate, call total refusal zero error, or infer a mechanism tag from static observational equality? The finite kernel makes those errors structurally unavailable within its declared data types.

For frames (a,b,c), a translation (T_{ab}) acts on a point (p) by addition of an explicit displacement vector. Composition is therefore

$$T_{bc}(T_{ab}(p)) = T_{ac}(p), \quad T_{ac} = T_{ab} + T_{bc}. \quad (54)$$

The inverse carries the negative displacement and satisfies

$$T_{ba}^{-1}(T_{ab}(p)) = p. \quad (55)$$

These are finite translation theorems, not a claim that every biological coordinate transformation is translational. Their purpose is to require an explicit transform witness before a cross-frame distance is evaluated.

An accepted architecture estimate contains nonnegative integer weights for local, broadcast, and recurrent contributions together with a positive total S :

$$w_L + w_B + w_R = S, \quad S > 0, \quad 0 \leq w_k \leq S. \quad (56)$$

The executable estimators use real-valued simplex projections; the integer kernel verifies the corresponding finite bookkeeping invariant without claiming that the integer representation is the estimator itself.

Selective error is a partial quantity. For accepted count (n_A) and error count ($n_E \leq n_A$), the formal interface returns

$$\text{SelErr}(n_E, n_A) = \begin{cases} \text{undefined}, & n_A = 0, \\ (n_E, n_A), & n_A > 0. \end{cases} \quad (57)$$

Thus an all-refusal condition cannot masquerade as perfect accuracy. The theorem preserves the exact count pair rather than formalizing floating-point division or a statistical confidence interval.

Finally, let m_L, m_B, m_R be models whose mechanism tags are pairwise distinct. Constant-output constructions provide an explicit witness such that

$$\begin{aligned} \text{tag}(m_L) &\neq \text{tag}(m_B), & \text{tag}(m_L) &\neq \text{tag}(m_R), \\ \text{tag}(m_B) &\neq \text{tag}(m_R), & \forall y, & m_L(y) = m_B(y) = m_R(y). \end{aligned} \quad (58)$$

Equation 58 is constructive: static observational equality does not logically imply equality of mechanism tags when the model families are unrestricted. It formalizes the paper’s reason for requiring temporal transfer, interventions, nuisance controls, and restricted model classes. The kernel does not prove that any biological population is local, broadcast, recurrent, or gravitationally organized.

12. Falsifiers and decision rules

The paper’s architecture claims are useful only if they can fail.

12.1 Evidence against a strong local-only model

A strong local-only model would be weakened if focal perturbation of a compact source caused rapid, object-general, cross-regional shifts that remained after measured common inputs and behavior were controlled, especially if independent receiver perturbations had little reciprocal effect.

12.2 Evidence against a strong single-broadcast model

A strong single-broadcast model would be weakened if no privileged source could be found across tasks, perturbations had reciprocal and path-dependent effects, settling time scaled with network distance, and partial estimates survived source perturbation through distributed compensation.

12.3 Evidence against a strong recurrent-distributed model

A strong recurrent-distributed model would be weakened if a low-dimensional source signal explained receiver estimates, perturbation effects were rapid and source-asymmetric, recurrence-specific timing

added no held-out predictive value, and disconnection of receiver-receiver pathways did not affect estimation beyond lost feedforward input.

12.4 Evidence against the need for an explicit gravity estimate

The broader framing would be weakened if flexible behavior, neural transfer, and adaptation were better predicted by coordinate-free policies that never require a recoverable gravity-related latent variable. In that case, "gravitational reference frame" may be a useful analyst description rather than an explicitly represented neural quantity.

12.5 Evidence boundary for SAN integration

The PWD bridge would be weakened if receiver-relative timing or state departures added no reproducible predictive or causal information after firing rate, stimulus, behavior, and ordinary network state were modeled. The failure would be recorded at the named preparation, variables, and endpoint and used to refine the measurable operator. It would not justify replacing SAN's full source-faithful account with a weaker phase-only strawman.

13. Limitations

First, most of the central neural evidence comes from macaque tasks involving controlled objects, scenes, and tilt. Generalization to natural behavior, humans, and conscious experience remains open.

Second, the real-data reproduction uses released summary arrays rather than raw spike trains. It can verify category counts but cannot re-estimate all preprocessing choices or test network timing.

Third, both synthetic benchmarks are intentionally idealized. Perfect held-out classification in the first benchmark reflects separated pure generators and fixed simulation assumptions. The second benchmark adds mixtures and selected shifts, but still uses one ring topology, six handcrafted features, a centroid-mixture estimator, and known synthetic ground truth. Neither result is a biological performance claim.

Fourth, gravity-aligned responses can arise from visual regularities, vestibular and somatosensory cues, learned priors, or their combination. A measured "gravity frame" must therefore be operationally tied to cue-conflict and intervention conditions.

Fifth, cortex is unlikely to implement one pure architecture. The mixed-architecture benchmark demonstrates one auditable mixture-and-unresolved contract, but a biological analysis must estimate uncertainty over mixtures and identify when architecture changes with task, learning, brain state, hidden nodes, or common drive.

Sixth, the term "cortical network" does not imply that subcortical structures are secondary. Vestibular organs, brainstem pathways, thalamic relays, cerebellar computation, basal ganglia, and body feedback may be essential components of the complete loop.

Finally, the historical genealogy establishes development of the author's proposal, not independent empirical validation. The empirical status rests on primary experiments, executable reproductions, and future discriminating tests.

The protocol-and-power analysis adds planning scenarios but no prospective pilot variance, attrition estimate, institution-specific feasibility constraint, or ethics determination. The reported power crossings show how assumptions interact; they are not fixed sample sizes. Small independent-unit counts can cross a power threshold for large assumed effects while still producing wide uncertainty and unstable generalization.

The registered missingness evaluation expands missingness and hidden-node coverage but remains a frozen synthetic evaluation of one ring-family generator and one six-feature estimator. Its intervals quantify repeated synthetic rows, not biological population sampling. The conditions are declared stressors rather than a complete model of electrode drift, neural nonstationarity, behavior-dependent missingness, anesthesia, lesion compensation, or cross-subject heterogeneity. The strong nominal result and the adverse combined-confound result are equally local to this generator family. A future study must add alternative graph topologies, stronger estimators, matched baselines, and generator-family shifts without selecting them after seeing the final test.

The topology-and-estimator evaluation adds ring, line, small-world, and modular graphs and two stronger multi-topology estimators, but all four estimators still operate on handcrafted summaries of a shared generator family. The nearest-neighbor model can recover exact development-supported mixtures in nominal synthetic data; that is not evidence that nearest neighbors approximate cortical inference. Measurement noise, hidden common input, the unsuccessful control transform, and the combined shift materially reduce accuracy or coverage. Selective-risk calibration was performed on synthetic seeds from the same declared generator family and may fail under a different family. The feature ablation does not establish biological necessity: timing and spread matter within the frozen benchmark, while dispersion is not uniquely identified by the registered endpoints. Independent generator families, learned time-series models, topology uncertainty, and prospective biological data remain necessary.

The independent-generator time-series evaluation supplies one independently coded generator and learned temporal readouts, but it is still synthetic. The fixed reservoir is not a faithful biological circuit or a full recurrent neural-network baseline; only its readout is learned. The topology-conditioned estimator is given the true synthetic topology label and trained on the directed hierarchy that challenges the topology-marginal model. The proxy-adjusted model is developed on common-cause examples, so its favorable common-cause result measures augmentation plus proxy use rather than an unanticipated causal discovery. A single proxy is inadequate under heterogeneous node loadings, and combined shift exceeds the calibration risk target. The intervention operators remove generator components by construction; they validate bookkeeping and expected sensitivity, not experimental causal identification. Independent implementation reduces one reuse risk but does not create independence from all modeling assumptions.

The bounded formal-contract verification verifies fourteen propositions only in the finite Lean structures declared by the proof source. Translation is modeled by integer displacement, architecture weights by nonnegative integers with a positive total, selective error by an optional count pair, and static nonidentifiability by unrestricted constant-output witnesses. The kernel does not prove empirical transform validity, floating-point estimator correctness, cryptographic provenance, intervention uptake, biological architecture, gravitational physics, neural rendering, conscious content, or SAN as a complete theory. Those obligations remain in the empirical, executable, and interpretive layers of the program.

The capacity-matched learned-view evaluation remains a synthetic observation-view comparison. Its four recurrent cores have equal parameter counts, initialization, optimizer, data allocation, and

endpoint rules, but their engineered views differ in information content and inductive bias. Equal parameter count does not imply equal effective complexity, optimization difficulty, or biological fidelity. The learned comparison uses one initialization and one frozen generator family. The historical fixed reservoir is a useful descriptive control but is not capacity matched and is not part of the primary multiplicity family. Seed-cluster intervals quantify repeated synthetic seeds, not animals or people. The final biological test remains sealed.

14. Conclusion

The nervous system can express object and scene relations in gravity-aligned, retinal, vestibular, body-centered, and hybrid coordinates. Those representations do not disclose the complete architecture that produced them. Local estimation, specialist broadcast, and recurrent-distributed constraint can remain observationally equivalent in static data, so the decisive evidence must join time, topology, nuisance measurement, perturbation, adaptation, transfer, and an allowed unresolved state.

Released electrophysiology reproduces the bounded gravity/retina partitions but cannot select an architecture. Successively stronger synthetic studies show both the promise and the fragility of the proposed discriminator. Pure and mixed endpoints can be recovered under declared nominal conditions; missing causal channels trigger refusal; hidden common drive, topology shift, node loss, and combined distribution shift create large failure surfaces. A finite Lean model verifies the reporting contract and constructive static nonidentifiability, not a biological mechanism.

The capacity-matched learned-view study sharpens the final pre-biological result. Graph-organized measurements have the lowest registered aggregate error among four equal-parameter recurrent views, yet the historical fixed reservoir has descriptively lower error with far lower shifted coverage, the held-out directed hierarchy reverses the graph-versus-broadcast ordering, and interventions retain structural nulls and reversals. The defensible result is therefore a falsifiable measurement and experiment program, not a cortical verdict. The biological final test remains unopened.

Data, code, and reproducibility statement

The reproducibility companion contains the exact released arrays and custody record; pure, mixed, missingness, topology, independent-generator, and learned-view benchmarks; frozen contracts and pre-outcome commitments; raw and summary tables; capacity-matched checkpoints; formal Lean source and compiler receipts; all adverse, null, refused, unstable, topology-specific, nuisance-sensitive, intervention, and disagreement results; and exact checksums.

The capacity-matched evaluation contributes 37,632 primary predictions, 5,376 intervention rows, four 291-parameter checkpoints, 480 epoch records, three figures, paired-seed contrasts, and the seed-cluster uncertainty tables. Recomputing uncertainty at the twenty-four-seed cluster level changes no point estimate, endpoint, ranking, raw output, primary digest, or intervention digest. The primary and intervention digests remain 3C151F729212C73A52DADC1E76E3B8696C7EDA53E0E50270BDA3712BD261FF0E and A9528444D72F4F9B29C30E997FB6967050486B428CBAF19F30AE1806E0C1330.

The biological final test is sealed and was not opened for this preprint.

Ethics statement

No new human or animal experiment was performed. The biological studies cited here were conducted under their original approvals. The prospective experiments require independent scientific, welfare, ethics, and institutional review before execution.

Conflicts of interest

The author declares no financial conflict of interest. Self-Aware Networks is the author's theoretical program; that relationship is stated so readers can distinguish source genealogy from independent empirical evidence.

References

1. Emonds, A., Srinath, R., Nielsen, K. J., & Connor, C. E. (2023). Object representation in a gravitational reference frame. *eLife*, *12*, e81701. <https://doi.org/10.7554/eLife.81701>
2. Rosenberg, A., & Angelaki, D. E. (2014). Gravity influences the visual representation of object tilt in parietal cortex. *Journal of Neuroscience*, *34*, 14170-14180. <https://doi.org/10.1523/JNEUROSCI.2030-14.2014>
3. Vaziri, S., & Connor, C. E. (2016). Representation of gravity-aligned scene structure in ventral pathway visual cortex. *Current Biology*, *26*, 766-774. <https://doi.org/10.1016/j.cub.2016.01.022>
4. Chen, X., DeAngelis, G. C., & Angelaki, D. E. (2013). Diverse spatial reference frames of vestibular signals in parietal cortex. *Neuron*, *80*, 1310-1321. <https://doi.org/10.1016/j.neuron.2013.09.006>
5. Caruso, V. C., Pages, D. S., Sommer, M. A., & Groh, J. M. (2021). Compensating for a shifting world: evolving reference frames of visual and auditory signals across three multimodal brain areas. *Journal of Neurophysiology*, *126*, 82-94. <https://doi.org/10.1152/jn.00385.2020>
6. Pramod, R. T., Cohen, M. A., Tenenbaum, J. B., & Kanwisher, N. (2022). Invariant representation of physical stability in the human brain. *eLife*, *11*, e71736. <https://doi.org/10.7554/eLife.71736>
7. Deneve, S., Latham, P. E., & Pouget, A. (2001). Efficient computation and cue integration with noisy population codes. *Nature Neuroscience*, *4*, 826-831. <https://doi.org/10.1038/90541>
8. Pouget, A., Deneve, S., & Duhamel, J.-R. (2002). A computational perspective on the neural basis of multisensory spatial representations. *Nature Reviews Neuroscience*, *3*, 741-747. <https://doi.org/10.1038/nrn914>
9. Fetsch, C. R., Turner, A. H., DeAngelis, G. C., & Angelaki, D. E. (2009). Dynamic reweighting of visual and vestibular cues during self-motion perception. *Journal of Neuroscience*, *29*, 15601-15612. <https://doi.org/10.1523/JNEUROSCI.2574-09.2009>
10. Attneave, F., & Reid, K. W. (1968). Voluntary control of frame of reference and slope equivalence under head rotation. *Journal of Experimental Psychology*, *78*, 153-159. <https://doi.org/10.1037/h0026150>

11. Baier, B., Thomke, F., Wilting, J., Heinze, C., Geber, C., & Dieterich, M. (2012). A pathway in the brainstem for roll-tilt of the subjective visual vertical: evidence from a lesion-behavior mapping study. *Journal of Neuroscience*, *32*, 14854-14858. <https://doi.org/10.1523/JNEUROSCI.0770-12.2012>
12. Brandt, T., Dieterich, M., & Danek, A. (1994). Vestibular cortex lesions affect the perception of verticality. *Annals of Neurology*, *35*, 403-412. <https://doi.org/10.1002/ana.410350406>
13. Bock, O., & Dalecki, M. (2015). Mental rotation of letters, body parts and scenes during whole-body tilt. *Human Movement Science*, *40*, 352-358. <https://doi.org/10.1016/j.humov.2015.01.017>
14. Connor, C. E., & Knierim, J. J. (2017). Integration of objects and space in perception and memory. *Nature Neuroscience*, *20*, 1493-1503. <https://doi.org/10.1038/nn.4657>
15. Shepard, R. N., & Metzler, J. (1971). Mental rotation of three-dimensional objects. *Science*, *171*, 701-703. <https://doi.org/10.1126/science.171.3972.701>
16. Laurens, J., & Angelaki, D. E. (2018). The brain compass: a perspective on how self-motion updates the head direction cell attractor. *Neuron*, *97*, 275-289. <https://doi.org/10.1016/j.neuron.2017.12.020>
17. Hawkins, J., Lewis, M., Klukas, M., Purdy, S., & Ahmad, S. (2019). A framework for intelligence and cortical function based on grid cells in the neocortex. *Frontiers in Neural Circuits*, *12*, 121. <https://doi.org/10.3389/fncir.2018.00121>
18. Felleman, D. J., & Van Essen, D. C. (1991). Distributed hierarchical processing in the primate cerebral cortex. *Cerebral Cortex*, *1*, 1-47. <https://doi.org/10.1093/cercor/1.1.1-a>
19. Blumberg, M. (2011/2022). Stereographic distributed-pixel and multisensory prediction note (a0348z.md). Owner date May 13, 2011; public Git fixation June 8, 2022. <https://github.com/v5ma/selfawarenetworks/blob/main/a0348z.md>
20. Blumberg, M. (2012). *Inner Peace* [public recording]. <https://www.youtube.com/watch?v=QX-0ef31tAU>
21. Blumberg, M. (2017). *Neural Lace Podcast: Spatial Computing and Artificial Neurology* [public audio and transcript]. https://youtu.be/YrX_68oKuVs
22. Blumberg, M. (2024). *Self Aware Networks*, version 9, gravitational-reference-frame discussion. Public Git fixation January 4, 2024. <https://github.com/v5ma/selfawarenetworks/blob/main/SA0v9.md>
23. Blumberg, M. (2024). Cortical dynamics and neurocircuitry: mapping sensory loops and cognitive pathways. *Silicon Valley Global News*. <https://www.svgn.io/wiki/svgn/articles/2024-01-24-cortical-dynamics-and-neurocircuitry-mapping-sensory-loops-and-cognitive-pathways-svgn-140981414>
24. Elman, J. L. (1990). Finding structure in time. *Cognitive Science*, *14*, 179-211. https://doi.org/10.1207/s15516709cog1402_1
25. Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. *Nature*, *323*, 533-536. <https://doi.org/10.1038/323533a0>
26. Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. *International Conference on Learning Representations*. <https://arxiv.org/abs/1412.6980>